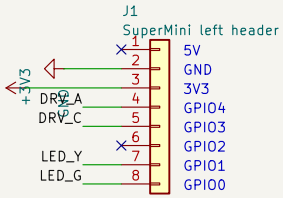
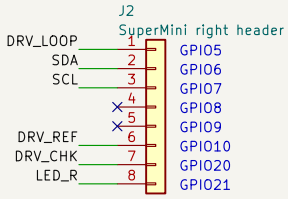
Dermasense minimum two-zone reader v0.1 (bench, PROVISIONAL)

Bipolar DC drive per zone; values from 6_electronics/calc_readout.py. USB-powered bench tool, not for patient use.

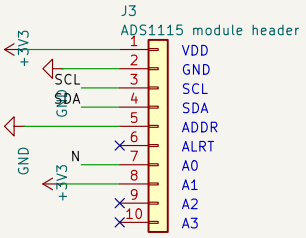
U1 ESP32-C3 SuperMini (module headers)





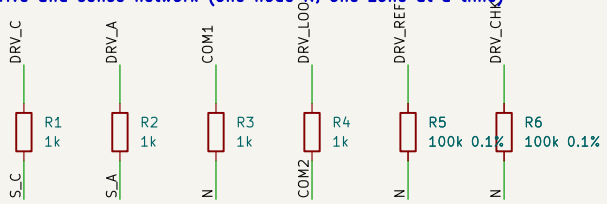
GPIO2/8/9 = strapping pins (8 = on-board LED, 9 = BOOT): left unused. 5V pin unused (USB powers the module).

U2 ADS1115 ADC module



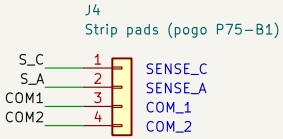
I2C address 0x48 (ADDR to GND); 10k pull-ups are on the module.
A0 = node N (zone/loop/check). A1 = 3V3 rail (ratiometric Vdd).

Drive and sense network (one node N, one zone at a time)





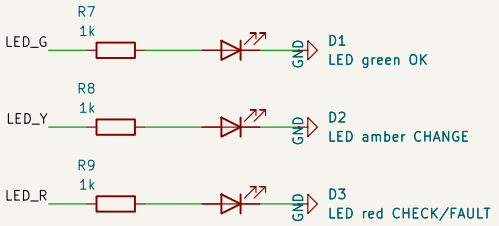
J4 strip contact: 4 pogo pins in clamp_base_v0_1



Power: USB-C -> SuperMini LDO -> 3V3



Status LEDs (1.3 mA each)



Measurement sequence (firmware 7_firmware/reader_v0_1):
1 Zone C: DRV_C and DRV_REF drive, all other drive pins high-Z. Phase +: DRV_C=1, DRV_REF=0; phase -: DRV_C=0, DRV_REF=1.
Rx = Rref(Vdd/VN - 1) (phase +), Rx = Rref VN/(Vdd - VN) (phase -); average; subtract R1 + R3 = 2 x 1k.
2 Zone A: same with DRV_A. 3 Loop: DRV_LOOP vs DRV_REF through COM2 -> COM loop -> COM1 (open = FAULT).
4 Check: DRV_CHK vs DRV_REF (known 100k) -> gain/loading correction. Between scans every drive pin is high-Z (no DC on the strip).
Thresholds (calc_readout): wet < 100 kohm, dry > 300 kohm, 3 consecutive scans. States: OK / CHANGE / CHECK (A wet before C) / FAULT.

Generated by 6_electronics/kicad/make_reader_schematic.py - edit the script, not this file
Bench reader: ESP32-C3 SuperMini + ADS1115 module
Team Dermasense, Vishwakarma Awards 2026-27

Sheet: /
File: reader_v0_1.kicad_sch

Title: Dermasense two-zone reader v0.1

Size: A4	Date: 2026-10-14	**Rev: 0.1 PROVISIONAL**
KiCad E.D.A. 10.0.6		Id: 1/1